

Computer Architecture Lab (Week 3)

Decimal Number

- ☐ Converting decimal number to binary number and hexadecimal number
- ☐ 57
- ☐ 143
- ☐ 76
- ☐ 223
- ☐ 90

Hexadecimal Number

- ☐ Converting hexadecimal number to binary number and decimal number
- ☐ 57H
- ☐ 143H
- ☐ 76H
- ☐ 223H
- ☐ 90H

Binary Number

- ☐ Converting binary number to decimal number and hexadecimal number
- ☐ 11111100
- ☐ 00111100
- ☐ 1111000000001111
- ☐ 10101100
- ☐ 0011001100001111

Opcode	Operand	Description
RAR	None	Rotate accumulator right through carry

- Each binary bit of the accumulator is rotated right by one position through the Carry flag.
- Bit D0 is placed in the Carry flag, and the Carry flag is placed in the most significant position D7.
- CY is modified according to bit D0.
- S, Z, P, AC are not affected.
- **Example:** RAR.

Opcode	Operand	Description
RAL	None	Rotate accumulator left through carry

- Each binary bit of the accumulator is rotated left by one position through the Carry flag.
- Bit D7 is placed in the Carry flag, and the Carry flag is placed in the least significant position D0.
- CY is modified according to bit D7.
- S, Z, P, AC are not affected.
- **Example:** RAL.

Opcode	Operand	Description
RLC	None	Rotate accumulator left

- Each binary bit of the accumulator is rotated left by one position.
- Bit D7 is placed in the position of D0 as well as in the Carry flag.
- CY is modified according to bit D7.
- S, Z, P, AC are not affected.
- **Example:** RLC.

Opcode	Operand	Description
RRC	None	Rotate accumulator right

- Each binary bit of the accumulator is rotated right by one position.
- Bit Do is placed in the position of D7 as well as in the Carry flag.
- CY is modified according to bit Do.
- S, Z, P, AC are not affected.
- **Example:** RRC.

Arithmetic Instructions

Opcode	Operand	Description
INR	R M	Increment register or memory by 1

- The contents of register or memory location are incremented by 1.
- The result is stored in the same place.
- If the operand is a memory location, its address is specified by the contents of H-L pair.
- **Example:** INR B or INR M

Arithmetic Instructions

Opcode	Operand	Description
INX	R	Increment register pair by 1

- The contents of register pair are incremented by 1.
- The result is stored in the same place.
- **Example:** INX H

Jump Conditionally

Opcode	Description	Status Flags
JC	Jump if Carry	CY = 1
JNC	Jump if No Carry	CY = 0
JP	Jump if Positive	S = 0
JM	Jump if Minus	S = 1
JZ	Jump if Zero	Z = 1
JNZ	Jump if No Zero	Z = 0
JPE	Jump if Parity Even	P = 1
JPO	Jump if Parity Odd	P = 0

Data Transfer Instructions

Opcode	Operand	Description
PUSH	Reg. pair	Push register pair onto stack

- The contents of register pair are copied onto stack.
- SP is decremented and the contents of high-order registers (B, D, H, A) are copied into stack.
- SP is again decremented and the contents of low-order registers (C, E, L, Flags) are copied into stack.
- **Example:** PUSH B

Data Transfer Instructions

Opcode	Operand	Description
POP	Reg. pair	Pop stack to register pair

- The contents of top of stack are copied into register pair.
- The contents of location pointed out by SP are copied to the low-order register (C, E, L, Flags).
- SP is incremented and the contents of location are copied to the high-order register (B, D, H, A).
- **Example:** POP H

Flage Register in 8085 Microprocessor

- ❑ What is Flage in register in 8085 microprocessor?
- ❖ Flag register includes five flip-flops, which are set or reset after an operation according to the data conditions of the result in the accumulator and other registers.
- ❖ The flags register can have a total of eight flags. Thus, a flag can be represented by 1 bit of information. But only five flags are implemented in 8085.

D7	D6	D5	D4	D3	D2	D1	D0
S	Z	X	AC	X	P	X	CY

Flage Register in 8085 Microprocessor

☐ Z (Zero) Flag :

- ❖ This flag indicates whether the result of mathematical or logical operation is zero or not.
- ❖ If the result of the current operation is zero, then this flag will be set, otherwise reset

☐ CY (Carry) Flag :

- ❖ This flag indicates, whether, during an addition or subtraction operation, carry or borrow is generated or not, if generated then this flag bit will be set.

Flage Register in 8085 Microprocessor

❑ AC (Auxiliary Carry) Flag:

- ❖ It shows carry propagation from D3 position to D4 position.
- ❖ AF (Auxiliary carry Flag): It holds the carry (half-carry) after addition or borrows after subtraction between bit positions 3 and 4 of the result in a BCD operation.
- ❖ If carry or borrow generated to D3 and to pass to D4, then AC flag is set otherwise reset.

❑ PF (Parity Flag):

- ❖ It is the count of ones in a number expressed as even or odd.
- ❖ It is logic 0 for even parity (i.e. even number of 1s) and logic 1 for odd parity (i.e. odd number of 1s).
- ❖ For example, if a number contains three binary one bits, it has odd parity and if a number contains no one bits, it has even parity.

Flage Register in 8085 Microprocessor

❑ SF (Sign Flag):.

- ❖ It holds the arithmetic sign of the result after arithmetic or logic instruction executes.
- ❖ If $S = 1$, the sign bit (leftmost bit of a number) is set or negative and if $S = 0$, the sign bit is cleared or positive.

TP01: Right shift 4 bits of 8-bits data (shift right with carry)

- ❑ **Statement:** Do 4 bits shifting of 8-bits data in memory location 2000H and store shifting data to memory location 2001H.
- ❑ **Sample Problem:** (2000H) = 5B = 0101 1011 (CY = 0)
- ❑ **Result:** (2001H) = 65 = 0110 0101 (CY = 1)

TP02: Right shift 8 bits of 16-bits data (shift right without carry)

- ❑ **Statement:** Do 8 bits shifting of 16-bits data in memory location 2000H and 2001H then store shifting data to memory location 2002H and 2003H.
- ❑ **Sample Problem:** (2001H & 2000H) = 04 05 = 0000 0100 0000 0101
- ❑ **Result:** (2003H & 2002H) = 05 04 = 0000 0101 0000 0100

TP03: Left shift 4 bits of 8-bits data (shift left with carry)

- ❑ **Statement:** Do 4 bits shifting of 8-bits data in memory location 2000H and store shifting data to memory location 2001H.
- ❑ **Sample Problem:** $(2000H) = 16 = 0001\ 0110$, $CY = 0$
- ❑ **Result:** $(2001H) = 60 = 0110\ 0000$, $CY = 1$

TP04: Left shift 8 bits of 16-bits data (shift left without carry)

- ❑ Statement: Do 8 bits shifting of 16-bits data in memory location 2000H and 2001H then store shifting data to memory location 2002H and 2003H.
- ❑ Sample Problem: $(2001H \& 2000H) = 04\ 05 = 0000\ 0100\ 0000\ 0101$
- ❑ Result: $(2003H \& 2002H) = 05\ 04 = 0000\ 0101\ 0000\ 0100$

TP05: Alter the content of flags register in 8085

❑ Statement: Modify the content of all flags in 8085

❑ Sample Problem:

S	Z	X	AC	X	P	X	CY
1	0	1	1	0	1	0	0

❑ Result:

S	Z	X	AC	X	P	X	CY
0	1	0	0	1	0	1	1

TP06: Calculate the sum of series of 8 bits numbers (no carry)

- ❑ Statement: Calculate the sum of series of 8-bits data from memory location 2000H to 2003H using branching instruction to loop. Memory location 2010H will be store the amount of series number to sum. Then store the sum data to address 2005H.
- ❑ Sample Problem: $2000H = 05$, $2001H = 04H$
 $2002H = 50$, $2003H = 10H$
- ❑ Result: $(2005) = 69H$

TP07: Calculate the sum of series of 8 bits numbers (with carry)

- ❑ Statement: Calculate the sum of series of 8-bits data from memory location 2000H to 2003H using branching instruction to loop. Memory location 2010H will be store the amount of series number to sum. Then store the sum data to address 2005H.
- ❑ Sample Problem: $2000H = 5A$, $2001H = 10H$
 $2002H = 74$, $2003H = 85H$
- ❑ Result: $(2005) = 63H$, $(2006) = 01H$

TP08: Calculate the sum of $1+2+\dots+n$

- ❑ Statement: Calculate the sum from 1 to n number in memory location 2000H. Then store to address 2001H.
- ❑ Sample Problem: $(2000H) = 04H$
- ❑ Result: $(2001H) = 01 + 02 + 03 + 04 = 0A$

Thanks